

Title:

Behavioral Analysis of Employee Attrition (A Retention Strategy Case Study)

Executive summary:

This independent case study examines Employee Attrition using a synthetic dataset of **1,470 Employees**. The motto was to associate Workforce groups with higher attrition and determine interventions that can be used to test and improve retention.

The analysis compares attrition rates by Employee Satisfaction, Work-life balance, Overtime status and Commute distance. **The top 3 Strongest findings were:**

1. Fewer years at the company give rise to increased attrition
2. More commute distance leads to increased attrition.
3. More overtime status leads to increased attrition.

These results theorize that Employee departures may be related to elevated overtime status, Work-life balance and Job satisfaction levels. Furthermore, they are somewhat linked with perceived benefits like Income and Career value.

The recommended first action is Fix onboarding gaps. Based on a synthetic amount of **\$15,000 cost per departure**, a **10% reduction** in attrition could generate estimated savings of **\$300,000**. The findings given are observational and should be tested through, before full implementation.

DATA and METHOD:

Dataset: **IBM HR Analytics Employee Attrition & Performance.**

Sample: **1,470 records.**

Dataset type: **synthetic.**

Outcome: **Attrition Rate(%)**.

Variables analyzed: **overtime, satisfaction, work–life balance, tenure, commute, income.**

Method: **Comparison of Attrition rates.**

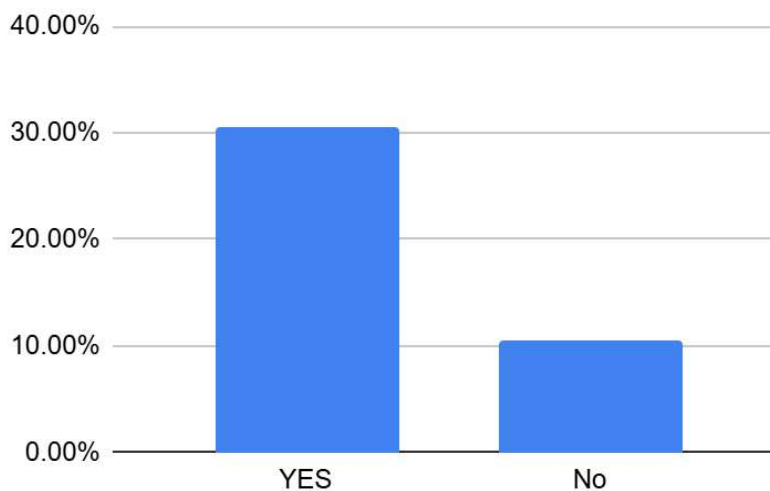
Limitation: —

Major Findings:

1. Findings 1:

Employees working overtime had an attrition rate of 30%, compared with 10% among employees not working overtime. This X-percentage-point difference is consistent with a workload-friction explanation: the daily effort and stress of overtime may increase the perceived cost of remaining in the role.

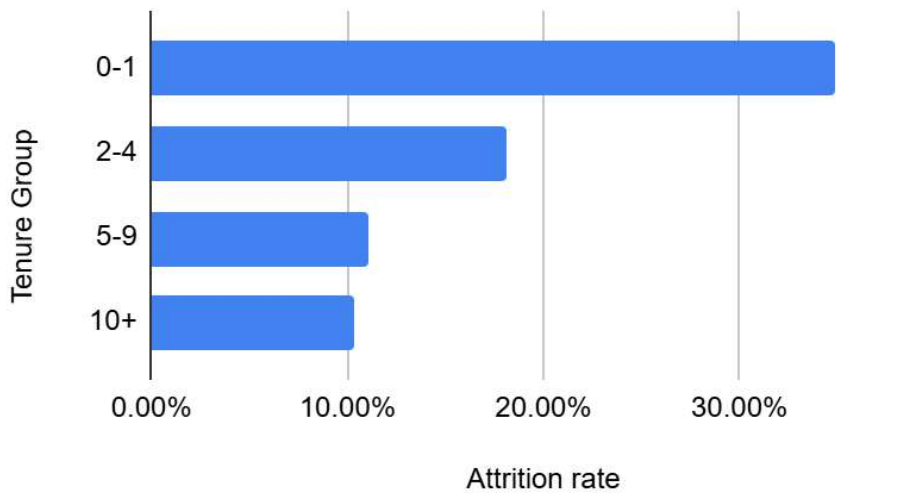
Employee Attrition Rate by Overtime...



2. Findings 2:

Employees in their first 0–1 years had an attrition rate of 35%. This could indicate gaps in onboarding, expectation-setting, belonging, or manager support.

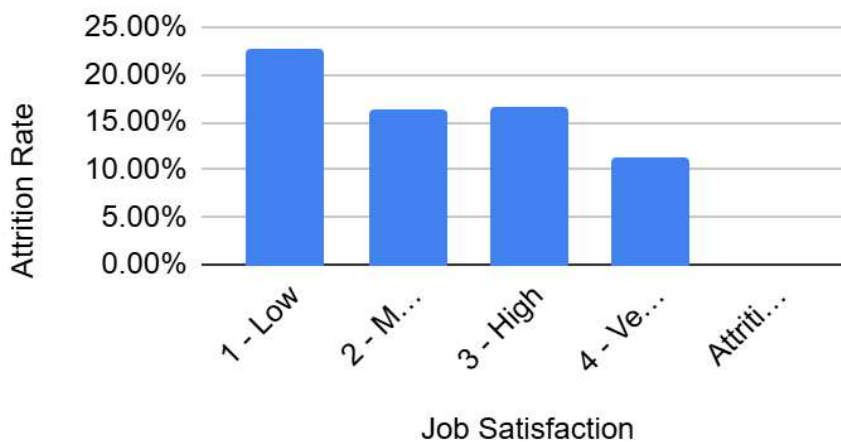
Employee Attrition Rate by Years at Com...



3. Findings 3:

Employees with low job satisfaction had an attrition rate of 22%, compared with 11% among those reporting very high job satisfaction. This suggests that daily job experience, recognition, progression, or role fit should be investigated.

Employee Attrition Rate by Job Sati...



RECOMMENDATIONS:

Priority	Recommendation	Behavioral mechanism	Success measure
1	Run workload reviews for teams with high overtime; introduce protected no-meeting blocks and clearer workload escalation	Reduces effort, fatigue, and work–life friction	Overtime hours, work–life balance, six-month attrition
2	Introduce 30-, 60-, and 90-day manager check-ins and a peer buddy for new employees	Increases clarity, belonging, support, and early commitment	First-year attrition, onboarding satisfaction, check-in completion
3	Give employees clear career pathways, internal-vacancy visibility, and structured development conversations	Increases perceived future value and reduces the appeal of leaving	Internal mobility, job satisfaction, attrition intention

I would promote a workload reduction pilot if Overtime demonstrates the highest attrition rates. To ensure a fair test, I suggest the pilot runs for 12 weeks in one department while another department works synonymously. The primary measures should be for overtime while secondary measures should be for Work-life balance over the next six months.

This analysis is eloquent and cannot authorize cause or effect. The synthetic dat set used lacks qualitative information like exit interview, manager quality, and personal circumstances. The proposed interventions should therefore be piloted and calculated using a comparison group if possible